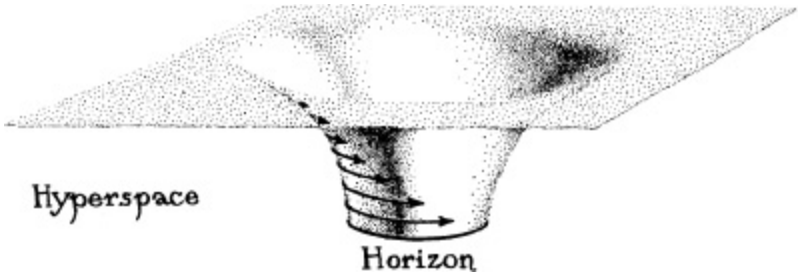
discovery was also made, independently, by Roger Penrose in London, and by Robert Boyer in Liverpool and Richard Lindquist, a former student of Wheeler's who was working at Wesleyan University in Middletown, Connecticut.) By the mid-1970s, Carter and others had gone on to show that Kerr's solution describes not just one special type of spinning black hole, but rather every spinning black hole that can possibly exist.

The physical properties of a spinning black hole are embodied in the mathematics of Kerr's solution, and Carter, by plumbing that mathematics, discovered just what those properties should be. One of the most interesting is a tornado-like swirl that the hole creates in the space around itself.



7.7 An embedding diagram showing the "tornado-like swirl" of space created by the spin of a black hole.

This swirl is depicted in the embedding diagram of Figure 7.7. The trumpet-horn-shaped surface is the hole's equatorial sheet (a two-dimensional piece of the hole's space), as embedded in a flat, three-dimensional hyperspace. The hole's spin grabs hold of its surrounding space (the trumpet-horn surface) and forces it to rotate in a tornado-like manner, with speeds proportional to the lengths of the arrows on the diagram. Far from a tornado's core the air rotates slowly, and, similarly, far from the hole's horizon space rotates slowly. Near the tornado's core the air rotates fast, and, similarly, near the horizon space rotates fast. At the horizon, space is locked tightly onto the horizon: It rotates at precisely the same rate as the horizon spins.

This swirl of space has an inexorable influence on the motions of particles that fall into the hole. Figure 7.8 shows the trajectories of two such particles, as viewed in the reference frame of a static, external observer—that is, in the frame of an observer who does not fall through the horizon and into the hole.

The first particle (Figure 7.Ba) is dropped gently into the hole. If the hole were not spinning, this particle, like the surface of an imploding star,